



CITY ENGINEER'S OFFICE
2101 O'Neil Avenue
Cheyenne, Wyoming 82001

A COMMUNITY OF CHOICE

MEMO

DATE: May 29, 2018
TO: Members of the Governing Body
FROM: Sam Berta, Staff Engineer

SUBJECT: Professional Services Agreement with DTS for Pavement Condition Survey.

The following is in support of why Staff has selected DTS for data collection services for the above project. DTS was determined as the most qualified firm for the Pavement Condition Survey Project. The selection process determined that DTS stood out as the best firm with experience, technical expertise and overall professional integrity giving the City and its residents the best product possible.

In the summer of 2018 the City will be looking to fully collect the condition of all the roadways in Cheyenne. In addition to road condition, curb and gutter condition will also be collected. The City has not completed a street condition survey since 2014. There is no existing curb and gutter condition information currently. The Pavement Condition Survey project will give the City of Cheyenne an opportunity to look at the existing road infrastructure and give the City a better picture on how to improve the overall pavement condition index (PCI) of the roads. The main intent to this project is to collect the current PCI of all the streets and condition of the curb and gutter. To address this need for data collection a plan was required. A Request for Proposal was drafted and advertised. In response to the RFP six proposals were submitted to the City. The engineering firms are as follows: DTS, PERTAN, SEH, IMS, Terracon and Black Pearl Asphalt.

After reviewing the six proposals submitted, three firms were selected for interviews. The top three firms that were selected are DTS, SEH and PERTAN. The interviews were conducted, and the scores were tallied, resulting in DTS being selected as the highest scoring firm. In summary, DTS scored higher than SEH and PERTAN in applicable experience, innovations, technical expertise, adequate resources, and equitable distribution of work. A unanimous decision was rendered with all of the selection committee choosing DTS.

It should be noted that Staff feels that DTS previous experience with street data collection and processing of many other communities was an aspect that the other firms were lacking. Also, their ability to collect and follow the ASTM D6433 standards with an automated system set them apart from the rest of the firms. This previous experience and knowledge would be advantageous

during the pavement condition survey project. Any problems experienced, and lessons learned could be taken into consideration during the data collection. This should result in fewer problems and more accurate data collected. Additionally, DTS's automated collection will be the safest means for the data collection.

At the conclusion of the interviews DTS was selected as the top firm for the project based on the established criteria. The estimated fees submitted by DTS were opened at that time and was found to be close to the anticipated amount. This amount was \$124,820.00.

In summary Staff recommends that the professional services agreement for the Pavement Condition Survey project be awarded to DTS in the amount of \$124,820.00.

City Contract # _____

PROFESSIONAL SERVICES AGREEMENT
for Pavement Condition Survey Project

PREAMBLE

THIS AGREEMENT is made and entered into this _____ day of _____, 2018, by and between the City of Cheyenne, Wyoming, a municipal corporation, located at 2101 O'Neil Avenue, Cheyenne, Wyoming 82001 ("City"), and Data Transfer Solutions, LLC, a limited liability company with an office located at 409 North Mason Ct., Suite 127, Fort Collins, Colorado 80524 ("Consultant"). Both the City and the Consultant may be referred to individually as "Party" or collectively as "Parties."

RECITALS

WHEREAS, the City desires professional engineering services in support of the Pavement Condition Survey Project;

WHEREAS, the Consultant is capable of performing the professional engineering services in support of the Pavement Condition Survey Project; and

WHEREAS, the Parties desire to memorialize the terms and conditions whereby the Consultant shall provide the needed professional engineering services to the City;

AGREEMENT

In consideration of the mutual covenants and obligations herein expressed, it is agreed by and between the Parties as follows:

1. **SCOPE OF WORK**

The Consultant agrees to provide services as described for the above-mentioned project, in accordance with the Scope of Work, attached hereto as Exhibit A, incorporated by reference and expressly made a part hereof.

2. **COMPENSATION**

In consideration of the services to be performed pursuant to this Agreement, the Consultant will bill the City and the City agrees to pay Consultant an amount not to exceed ONE-HUNDRED TWENTY-FOUR THOUSAND EIGHT-HUNDRED TWENTY DOLLARS AND NO CENTS (\$124,820.00), as stated in Exhibit B.

No additional funding under this Agreement shall be expended absent the Parties executing a modification pursuant to the terms of this Agreement. Any work performed in excess of the amount provided for herein and absent an executed modification under these terms shall not entitle the Consultant to any additional compensation.

3. TIME OF PERFORMANCE

The work included in this Scope of Work will be completed by no later than November 30, 2018 and in accordance with the schedule shown in Exhibit C.

4. CONSULTANT RESPONSIBILITY

In providing services under this Agreement, the Consultant shall perform in a manner consistent with that degree of care and skill ordinarily exercised by members of the same profession currently practicing under similar circumstances at the same time and in the same or similar locality. Upon notice to the Consultant and by mutual agreement between the Parties, the Consultant will, without additional compensation, correct those services not meeting such a standard.

5. COMPLIANCE WITH LAWS

The Parties agree that they will perform their obligations as provided in this Agreement in accordance with all applicable laws and ordinances.

6. DRUG-FREE WORKPLACE

In compliance with the Drug Free Work Place Act of November 1988, the City of Cheyenne has established an Alcohol and Controlled Substance Policy that pertains to alcohol and drug usage by City Employees. All architects, engineers and other consultants under contract with City of Cheyenne, and their employees and sub-consultants, are required to comply with the provisions of the City's Alcohol and Controlled Substance Policy for drug and/or alcohol usage on City property or other sites occupied by the Consultant while performing the duties and responsibilities of the Agreement. It is the responsibility of the Consultant to familiarize themselves with the requirements of this policy and to inform all their employees and sub-consultants of the requirements and ensure their compliance. If the Consultant, their employees or sub-consultants are found in violation of this policy, the Agreement may be terminated.

7. INDEMNIFICATION/HOLD HARMLESS

The Consultant agrees to indemnify, hold harmless and defend the City from and against any and all liabilities, claims, penalties, forfeitures and suits, and the cost and expenses incident thereto, including reasonable attorney's fees, which may hereafter arise as a result of death or bodily injury to any person, destruction or damage to any property, contamination of or adverse effects on the environment, or any violation of governmental laws, regulations or orders to the extent caused by (1) the Consultant's breach of any term or provision of this Agreement; or (2) any negligent or wrongful act, error or omission by the Consultant, or its employees or sub-consultants in the performance of this Agreement. The Consultant acknowledges that it may incur a financial obligation to the City pursuant to the terms of this paragraph.

8. INSURANCE REQUIREMENTS

The Consultant must provide proof of the following insurance coverages:

Commercial General Liability Insurance

For claims arising out of bodily injury, illness or death, or from damage to or destruction of property of others, including loss or use thereof, with minimum limits of \$1,000,000 per occurrence, \$2,000,000 aggregate for the entire term of the Agreement.

Business Automobile Insurance

Including owned, non-owned and hired vehicles with minimum limits for bodily injury and property damage of \$1,000,000 per accident for the entire term of the Agreement.

Workers' Compensation

Workers' compensation coverage shall be in effect for the entire term of the Agreement, as required by Wyoming law, for all employees or agents providing services under this Agreement. Consultant shall provide the City with proof of workers' compensation or employer's liability insurance coverage.

Professional Liability Insurance

The Consultant shall provide proof of professional liability insurance or errors and omissions liability insurance in an amount not less than \$500,000 to protect from any and all claims arising from the Consultant's negligence in the performance of duties under this Agreement. The City prefers that this liability insurance coverage be provided pursuant to an "occurrence" policy.

If this coverage is provided pursuant to a "claims made" policy:

- a. Consultant shall, concurrently with the execution of this Agreement, provide the City with a certificate of insurance demonstrating that such coverage is or shall be in effect at the time the Consultant begins the provision of services under this Agreement; and
- b. In the event the Consultant's services extend into a future policy period, the Consultant shall, prior to the policy expiration date, provide the City with a new certificate of insurance demonstrating that such coverage is or shall be in effect during all periods of time that Consultant will provide services under this Agreement; and
- c. Consultant shall maintain said "claims made" coverage for a period of five (5) years following the last date that Consultant has provided services under this Agreement; and
- d. In the event the Consultant or the insurer terminates "claims made" coverage prior to the expiration of the periods provided in subparagraphs a., b., or c. of this paragraph, the Consultant shall provide to the City advance written notification of the termination of said coverage and shall provide the City with an endorsement for an extended reporting period ("tail coverage") which shall be in effect for a period of time not less than three (3) years following the last date that Consultant has provided services under this Agreement.

Additional Insurance Information

The Consultant shall name the City of Cheyenne as an **Additional Insured** by

endorsement on its insurance policies, and shall provide the City with a copy of the endorsements. This requirement does not apply to workers' compensation and professional liability insurance policies.

Consultant shall provide the City with certificates of insurance acknowledging the above-stated coverages prior to beginning any work under this Agreement.

It is understood and agreed that these policies are primary and not contributory. All policies required under this Agreement shall be in effect for the duration of the Agreement. It shall be an affirmative obligation upon Consultant to immediately notify in writing the city risk manager, city clerk, and city attorney of any fact, circumstance, or occurrence that has resulted in or may result in the cancellation of, or substantive change to any insurance coverage required by this Agreement, and failure to do so shall be construed to be a breach of this Agreement.

If requested by the City, the Consultant shall provide the City with copies of insurance policy endorsements listing the City of Cheyenne as an additional insured. The City's failure to request or review such endorsements or certificates shall not affect the City's rights or Consultant's obligation hereunder.

Any insurance company providing coverage under this Agreement shall have a minimum A. M. Best rating of A- (excellent).

9. MINORITY AND DISADVANTAGED BUSINESS ENTERPRISES 49 C.F.R. Part 26
All Parties to this Agreement assure that no person will be excluded from participation in, denied the benefits of, or otherwise discriminated against, in connection with the award and performance of this Agreement on the grounds of age, race, color, disability, national origin or sex.
10. MONTHLY REPORT
With every monthly billing, Consultant shall provide the City Representative with a written statement of the status of the work with respect to the Scope of Work, time sheets, and work schedule. Failure to provide the required monthly report will delay processing of any payment request until the report is submitted.
11. INDEPENDENT CONSULTANT
The Consultant shall function as an independent consultant for the purposes of this Agreement. The Consultant shall assume sole responsibility for any debts or liabilities that may be incurred by the Consultant in fulfilling the terms of this Agreement. Nothing in this Agreement shall be interpreted as authorizing the Consultant or its agents or employees to act as an agent or representative of or on behalf of the City or to incur any obligation of any kind on the behalf of the City.
12. TAXES
The Consultant agrees to pay all valid taxes, excises, license fees, permit fees, bills, debts and obligations incurred by and in connection with its operations under this Agreement.
13. DEFAULT
Each and every term and condition in this Agreement shall be deemed to be a material element of the Agreement. In the event either party should fail or refuse to perform

according to the terms of this Agreement, such party may be declared in default.

14. REMEDIES

In the event a party has been declared in default, such defaulting party shall be allowed a period of fifteen (15) days within which to cure the default. In the event the default remains uncorrected, the non-defaulting party declaring default may elect to:

- a. Terminate the Agreement and seek damages, which damages shall not exceed the contract amount; or
- b. Treat the Agreement as continuing and require specific performance.

15. TERM AND TERMINATION

This Agreement shall be for the period stated in Exhibit C. The Parties may by mutual written agreement renew or extend this Agreement.

The City may, without cause, and upon thirty days' written notice to the Consultant, terminate this Agreement in whole or in part at any time, for the City's convenience. Upon receipt of such notice, the Consultant shall:

- a. Discontinue all services affected; and
- b. Deliver to the City within five (5) days all data, drawings, specifications, reports, estimates, summaries, and such other information and materials as may have been accumulated by the Consultant in performing this Agreement, whether completed or in process.
- c. In the event of termination for convenience, the City will pay the Consultant for accepted work done and expenses incurred to the date of termination. Such acceptance shall not be unreasonably withheld.

16. WAIVER

Failure by the City to specifically enforce any term shall not constitute waiver of such term. The waiver of any breach of any of the terms or conditions of this Agreement shall be limited to the act or acts constituting such breach, and shall never be construed as being a continuing or permanent waiver of any such terms or conditions, all of which shall be and remain in full force and effect as to future acts or happenings, notwithstanding any such individual waiver of any breach thereof.

The City's approval of drawings, plans, specifications, reports, and incidental work or materials furnished hereunder shall not in any way relieve the Consultant from responsibility for the technical accuracy of the work. The City's approval or acceptance of, or payment for, any services shall not be construed to operate as a waiver of any of the City's rights under this Agreement or any of its legal rights under statute and common law arising out of the performance of this Agreement.

17. SUCCESSORS AND ASSIGNS

All the terms, conditions, and provisions herein shall inure to the benefit of and be binding upon the Parties hereto and their respective permitted successors and assigns. Neither Party shall assign this Agreement without prior written consent of both Parties. Any delegation or assignment shall not operate to relieve either Party of its responsibilities hereunder.

However, nothing in this Agreement shall prohibit the Consultant from subcontracting to a qualified subcontractor.

18. THIRD-PARTY RIGHTS

The Parties do not intend to create in any other individual or entity the status of third party beneficiary, and this Agreement shall not be construed to create such status. The rights, duties, and obligations contained in this Agreement shall operate only between the Parties to this Agreement, and shall inure solely to the benefit of the Parties to this Agreement. The Parties to this Agreement intend and expressly agree that only the Parties signatory to this Agreement shall have any legal or equitable right to seek to enforce this Agreement, to seek any remedy arising out of a Party's performance or failure to perform any term or condition of this Agreement, or to bring action for breach of this Agreement. This paragraph is not intended nor shall it be construed to waive any of the Parties' immunities.

19. PERSONNEL

The Consultant agrees that it will employ at its own expense, all personnel necessary to perform the services required by this Agreement and in no event shall such personnel be employees of the City. All services required hereunder shall be performed by the Consultant and all personnel engaged therein shall be fully qualified under applicable federal, state and local law to undertake the work performed by them. The Consultant assumes full and sole responsibility for the payment of all compensation and expenses of such personnel and for all state and federal income tax, unemployment insurance, Social Security, disability insurance, and other applicable withholdings.

20. ADDITIONAL REMEDIES

In the event the Consultant fails to strictly perform in accordance with this Agreement, the City may elect to correct the deficiencies and charge the Consultant. In the event of default of any of the conditions by either party which shall require the party not in default to commence legal or equitable action against the defaulting party each party shall bear its own costs and expenses, including without limitation, attorney's fees.

21. GOVERNING LAW, JURISDICTION AND VENUE

The construction, interpretation and enforcement of this Agreement shall be governed by the laws of the State of Wyoming. The courts of the State of Wyoming shall have jurisdiction over any action arising out of this Agreement and over the Parties, and the venue shall be the First Judicial District, Laramie County, Wyoming.

22. GOVERNMENTAL IMMUNITY

The City of Cheyenne, along with its officials and employees, does not waive its governmental immunity by entering into this Agreement, except to the extent necessary for the Parties to pursue a contract action to clarify or enforce the written terms of the Agreement. Furthermore, the City specifically retain all immunities and defenses available to it as a sovereign or governmental entity pursuant to Wyo. Stat. § 1-39-101, *et seq.*, and all other relevant state and federal law. Designations of venue, choice of law, enforcement actions, and similar provisions should not be constructed as a waiver of governmental immunity. The Parties agree that any ambiguity in this Agreement shall not be strictly

construed, either against or for either party, except that any ambiguity as to governmental or sovereign immunity shall be construed in favor of governmental immunity.

23. MONITORING ACTIVITIES

The City shall have the right to monitor all activities related to this Agreement that are performed by the Consultant or its sub-consultants. This shall include, but not be limited to, the right to make site inspections at any time and with reasonable notice; to bring experts and consultants on site to examine or evaluate completed work or work in progress; to examine the books, ledgers, documents, papers, and records pertinent to this Agreement; and to observe personnel in every phase of performance of the related work.

24. NON-DISCRIMINATION

The Consultant shall comply with the Civil Rights Act of 1964, the Wyoming Fair Employment Practices Act (Wyo. Stat. § 27-9-105, *et seq.*), the Americans With Disabilities Act (ADA), as amended, 42 U.S.C. § 12101, *et seq.*, and the Age Discrimination Act of 1975 and any properly promulgated rules and regulations thereto, and shall not discriminate against any individual on the grounds of age, sex, creed, color, race, religion, national origin, ancestry, pregnancy or qualifying disability in connection with the performance under this Agreement.

25. PROFESSIONAL REGISTRATION

If applicable, the Consultant shall endorse, as required by law, plans and reports prepared under this Agreement, and shall affix thereto his or her seal of professional registration, showing that he or she is licensed to practice in the State of Wyoming, if necessary.

26. MODIFICATIONS

Any changes, modifications, revisions, or amendments to this Agreement which are mutually agreed upon by the Parties to this Agreement shall be incorporated by written instrument and executed by all Parties. No modification will be valid and binding but shall be void without a written instrument, executed by all Parties.

Any work performed by the Consultant that exceeds the scope of the Agreement without prior written execution of an instrument authorizing the work and demonstrating agreement by both Parties shall not be entitled to any additional compensation. Electronic communications including, but not limited to email or text messages, do not constitute written instruments sufficient to authorize such modifications. Executed instruments authorizing such modifications may be transmitted electronically.

27. INTELLECTUAL PROPERTY/WORK PRODUCT

All work (preliminary, draft, and final) produced by the Consultant under this Agreement is the property of the City. The City will own any and all data, documents, working papers, computer programs, photographs, forms, analyses, plans and other materials produced by the Consultant pursuant to this Agreement, and the Consultant hereby assigns and transfers to the City any and all intellectual property rights for such materials. The Consultant will provide the City with copies of all such materials including, without limitation, any research memoranda prepared under this Agreement.

Under no circumstances, including pending disputes between the City and the Consultant, will the Consultant fail to deliver possession of said documents and materials to the City upon demand. The Consultant agrees to indemnify, pay the defense costs of, and hold the City harmless from any and all claims, demands, costs, liabilities, losses, expenses and damages (including attorneys' fees, costs, and expert witnesses' fees) arising out of or in connection with this Agreement that sound in an intellectual property claim (including, but not limited to, patent, copyright, trademark, trade name, or trade secret infringement). This Section specifically survives the expiration or termination of this Agreement.

28. SEVERABILITY

If any provision of this Agreement is determined to be illegal or unenforceable for any reason, the same shall be severed from the Agreement and the remainder of the Agreement shall be given full force and effect

29. ENTIRETY OF AGREEMENT

This Agreement, consisting of nine (9) pages, Exhibit A, consisting of thirty-two (32) pages, Exhibit B, consisting of one (1) page, and Exhibit C, consisting of one (1) page, contains the entire understanding of the Parties. There are no other terms or conditions, written or oral, concerning or controlling this matter.

IN WITNESS WHEREOF, the Parties have hereunto set their hands and seals the day and year first above written.

SIGNATURE PAGE FOLLOWS

CITY OF CHEYENNE

(SEAL)

Attest:

By: _____

Marian J. Orr, Mayor

By: _____

Carol Intlekofer, City Clerk

Date: _____

CONSULTANT

By: _____

Print Name: _____

Date: _____

EXHIBIT A

Attached to Professional Services Agreement between the City of Cheyenne, Wyoming and Data Transfer Solutions, LLC for the Pavement Condition Survey Project

DTS Technical Proposal to City of Cheyenne RFP 20 – 18 – Pavement Condition Survey Project

years. Our knowledge of similar conditions found throughout Colorado, New Mexico, and Utah on previous projects provide us the background and confidence to correctly identify and evaluate these expected pavement conditions.

One unique capability that DTS offers the City of Cheyenne is the ability to take pavement condition survey samples at the same location each time that an assessment is conducted resulting in a more accurate analysis of the pavement performance for the City. DTS is performing this exact service for Bexar County, Texas as well as reviewing sampling methods versus 100% rating of the roads to determine the variance in accuracy obtained compared to the costing of complete rating of the roadway. As additional data is added to the database, the importance of repeatable data will be reflected in the accuracy and usefulness of the resulting deterioration curves. As previously recognized, the varying soil conditions can be more accurately modeled if the distress data is collected consistently and at the exact same areas to express the change in distress from the previous evaluation. DTS provides this accuracy and experience to understand how the next round of data will affect the deterioration models and the future budget projections.

4. Scope of Work

Statement of Project Requirements

DTS commits that our team, if selected and contracted, will perform the services outlined in Exhibit "A – Scope of Work" of the proposed Professional Services Agreement draft included in the RFP. DTS understands the goal of the project is to provide a pavement condition survey per the ASTM D6433 and that the City is utilizing MicroPAVER to manage its pavement management program. DTS will also provide an E70 file for consumption into PAVER as well as training on the pavement findings and utilization of MicroPAVER. Additionally, DTS will provide a pavement final report based on the City's maintenance and rehabilitation practices.

DTS is experienced with numerous data collection processes and has collected according to the American Society for Testing Materials (ASTM D6433-11) Standards of Practice established by the US Army Corps of Engineers for numerous agencies including Albuquerque, NM; Salt Lake City, UT; Pitkin County, CO; South Jordan, UT; El Paso, TX; Amarillo, TX; Houston, TX; Fort Worth, TX; San Antonio, TX; Indianapolis MPO, IN; Basalt, CO; Arlington, TX; Plano, TX; Virginia Beach, VA; Charleston County, SC; Orlando, FL; West Palm Beach, FL and Farmington, NM.

Inventory Verification, Image Capture and Inspection Data Collection

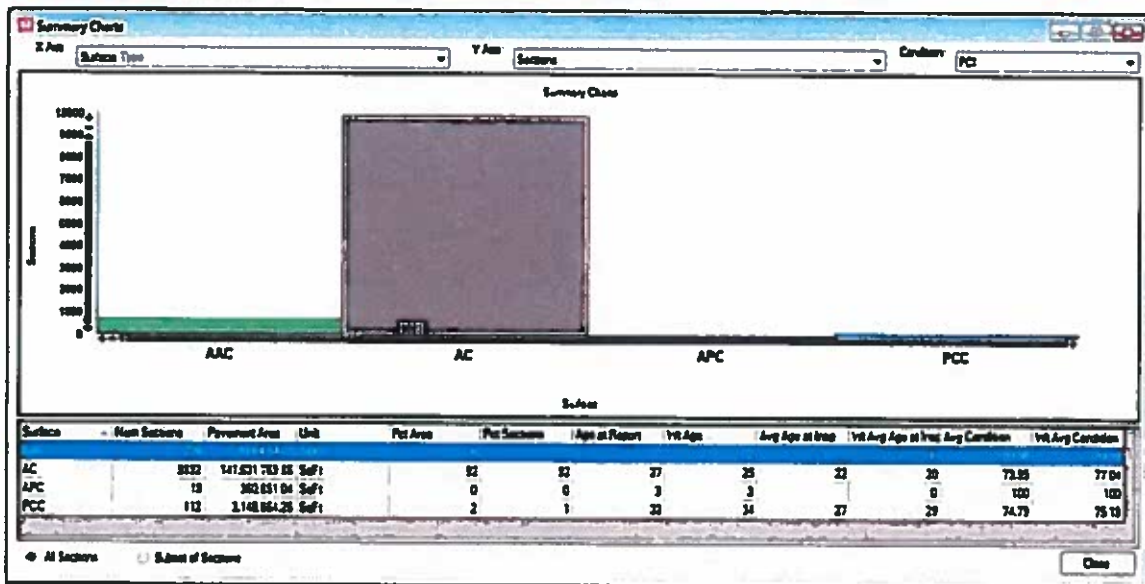
DTS understands the City maintains approximately 337 centerline miles of roadway and that it is using PAVER as its pavement management program. The PAVER database includes 940 Branches and 4,300 Sections. For this project, DTS will collect pavement data per the ASTM D6433/MicroPAVER criteria. We have vast experience with automated data collection and fully documented procedure, manual rating, imaging and distress extraction techniques, and we are working with the latest technology of automated collection to reduce the subjectivity of pavement condition rating. Our quality control includes numerous checks on image quality during collection and field validation with client staff to understand what the data is indicating in terms of pavement performance and deterioration. DTS will conduct the survey per the ASTM D6433 and identify the 20 asphalt and 19 concrete distresses.

DTS Technical Proposal to City of Cheyenne RFP 20 – 18 – Pavement Condition Survey Project

The DTS team has immediate availability with both its fleet of Mobile Asset Collection (MAC) vehicles and its staff to provide a timely and accurate pavement condition assessment for the City of Cheyenne.

DTS has unsurpassed experience with integration, configuration and review of the MicroPAVER database. A sample listing of clients where DTS has imported data and modeled pavement scenarios in MicroPAVER includes Orlando, FL; Indianapolis, IN; El Paso, TX; Mesquite, TX; Lewisville, TX; Seminole County, FL; Peoria, IL, Durham, NC and Pena Boulevard (CDOT). As a part of the City's Inventory Verification and Inspection Data Collection, DTS will utilize trained data collection/inspection teams to verify inventory data elements. DTS will collect the City's network with its fleet of Mobile Asset Collection (MAC) vehicles). Additionally, DTS team member Scott McDonald will first verify the inventory data present in the City's Paver database. Mr. McDonald will review City's PAVER database and work with the City to define all pavement distress types and treatment selections. Concurrent with the inventory Quality Control process, the data collection inspectors will collect inspection data for every section targeted by the City of Cheyenne for inspection (Current Paver database or specified segments thereto).

For instance, the DTS team will make every attempt to note any sections where there appears to be a significant discrepancy between what is observed and the Last Construction Date listed in the Section Inventory tab. This is a subjective call and would only be used if there was a concern with the current date offered by the City. In the above instance, this would be recommendations only, and the City staff would make the final judgement based on information provided.



MicroPAVER Summary of Pavement Type and Number of Segments Displayed in MicroPAVER

Our goal is to demonstrate to the City of Cheyenne that DTS employs an approach to mobile data collection and pavement management reporting that is repeatable and defensible across survey years, is GIS-centric at its core and is a quantitative approach that relies upon standards and procedures that can be applied for pavement management optimization. All data will be formatted to meet the ASTM

D6433-11 Standard Practice for Roads and Parking Lots Pavement Condition Index Surveys including International Roughness Index (IRI), rutting, cracking and faulting. DTS will provide all the required elements as requested by the City of Cheyenne and will provide a transparent collection and analysis process.

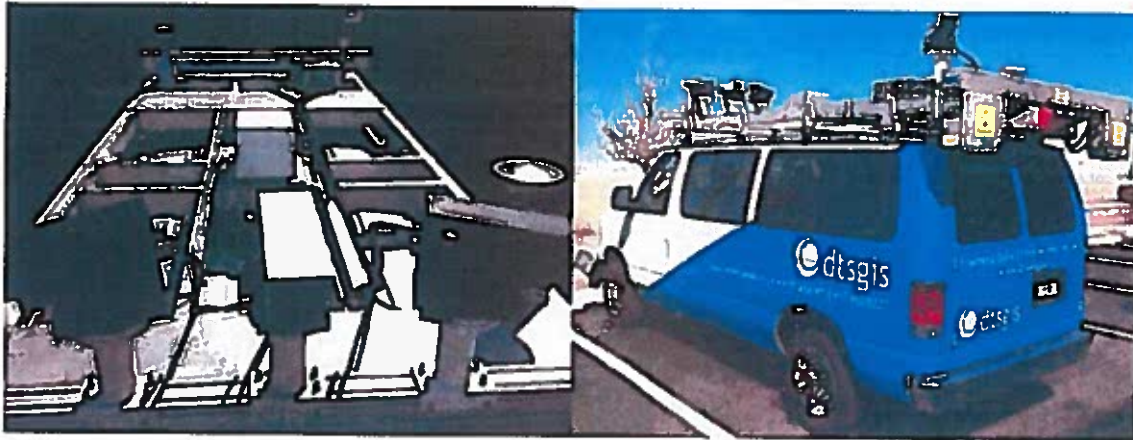
DTS will perform a complete physical pavement condition evaluation utilizing the MicroPAVER database sections identified by the City, and consistent with the rating conditions of MicroPAVER, and will inspect all directions: with samples approximately 100' in length for two-lane roadways, and 50' in length for wider roadways; will drive both directions of all non-divided roads; will drive both directions of any non-divided roads with four lanes or more, and roads that are divided shall be driven in both directions; will enter sample conditions into the MicroPAVER database; will create condition reports depicting current state in coordination with the City representative; will create a five-year plan based on the City's desired pavement condition rating and will provide recommendations and guidance on keeping the pavement management program updated.

DTS will collect Cheyenne's roadway inventory with 1 or more Mobile Asset Collection (MAC) vehicles. The collection routes will be calculated to minimize down times or areas of heavy congestion such as during Frontier Days or other festivals. DTS will work with the City of Cheyenne to set up a pilot project area so initial sample data can be collected and verified. DTS will collect data on the pilot project area and review the report results with City staff and acquire approval for full size network implementation. DTS proposes collecting pilot data in multiple locations to capture both asphalt and concrete surfaces for local roads.

Collection Method/Equipment

DTS collection vehicles are outfitted to collect color right-of-way images synchronized to the MAC DMI, IMU and GPS. The standard configuration of a MAC vehicle utilizes a 4-camera configuration: 3 forward-facing for a 120-degree view and one rear-facing for a 90-degree view. These cameras have a 1936x1456 resolution. Roadway photograph collection will be triggered by our DMI and will be calibrated to collect every 15 feet or more frequently if necessary based on the needs of the City. This will allow DTS to provide 100% continuous image coverage along the roadway. The forward cameras can auto-adjust from light to dark exposure and vice versa in 60 feet, quicker at lower speed situations. DTS will collect imagery during daylight hours only, with no rain, smoke, fog or snow visibility obstructions. DTS uses automatic shutter adjustments to allow for on-the-fly image brightening or darkening to avoid poor image quality from sunlight, overcast conditions or shadows. Any road segment that exhibits low image quality due to lighting will be recollected. DTS could also provide an inventory and condition assessment on roadway assets such as drop inlets, curb inlets, manholes, culverts, ditches, traffic signs, pavement markings/stripping, bridges, traffic signals and poles, sidewalks, curbs, trees, light poles, guardrails, ADA ramps and shoulders.

DTS Technical Proposal to City of Cheyenne RFP 20 – 18 – Pavement Condition Survey Project



A DTS Mobile Asset Collection (MAC) Vehicle

Each DTS Mobile Asset Collection vehicle is equipped with:

- **High-resolution right-of-way digital cameras** – Allied Vision Prosilica GX1920C GigE, frame rate of 15 images per second and 1936 x 1456 color resolution
- **Laser Road Imaging System (LRIS) pavement imaging system** – collects high-definition pavement images used to extract distress type severity and extent measurements. 4096 pixel/line, 28,000 lines/sec, 1mm resolution
- **ApplanixPOS220V inertial measuring unit (IMU)** – centimeter-level positioning of MAC van during collection
- **DMI equipment** – distance measuring instrument used for system integration
- **GPS equipment** – used for mapping level positioning of the vehicle, heading information and positional tagging of images. 2 positional units, 1 differential unit

DTS Technical Proposal to City of Cheyenne RFP 20 – 18 – Pavement Condition Survey Project

- **Servers** – on board servers for storing data, processing images and storing profiler, GPS, DMI and IMU data
- **Surface (road) profiler** – used for precise pavement ride and rut measurement

The MAC system collects all pavement and right-of-way images, IMU, DMI and profiler data concurrently. The vehicles will be configured to collect HD images of the road right-of-way while traveling at highway speeds. There is not a need for traffic control while the MAC vehicles are collecting ROW or pavement data since travel is conducted at posted highway speeds; however, our team will incorporate a Traffic and Safety Management Plan with an emphasis on the safety of the general public and our team members. As additional preventive measures, we include signage as well as a caution light alerting traffic of the MAC vehicles. Our data collection vehicles are serviced in intervals that exceed the manufacturer's recommendations. Additionally, all field crews deployed will have any required safety vests, hard hats or other equipment.

GPS Equipment

MAC vehicles utilize Applanix POS LV 420, delivering sub-meter accuracy of location data. The Applanix POS LV 420 is rated to 0.3 m accuracy in the X,Y position and 0.5 m in the Z position. The reference location for image collection system is the IMU mounted center of the MAC vehicle roof. The reference location for the MAC road profile unit is located on the left front corner of the profile unit. DMI, GPS data and images are synchronized by driving along a measured calibration run.

The DTS MAC fleet will collect right-of-way, profile and downward facing images in a single pass providing a complete view of the roadway, coupling images, inertial measurement, precise distance measurement and rigorous camera calibration that utilizes a "single heartbeat" to correlate all of these sensors into one single trajectory solution. This ensures a high level of accuracy and quality in the data.

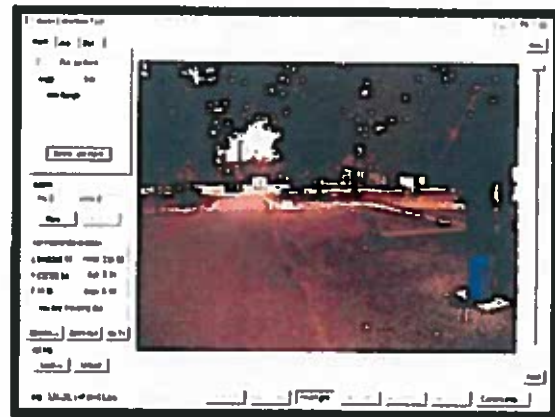
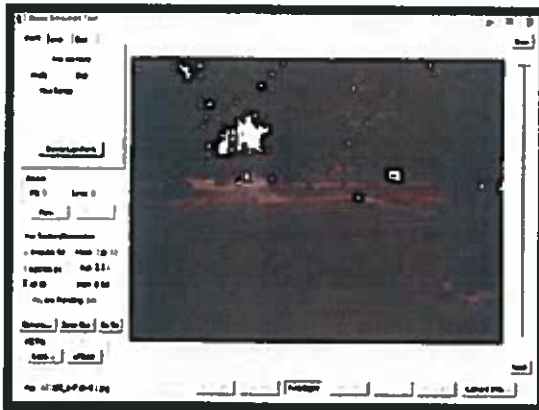
Additionally, our fleet of MAC vehicles is outfitted with the latest in industry collection equipment:

1. **Camera Resolution:** All images will be captured with digital output cameras with a resolution of 1936 x 1456 pixels.
2. **Image Compression and File Size:** Recorded images are compressed using the JPEG compression standard. Images are compressed in real-time during the time of capture. Image file size is typically around 150Kb per image and generally less than 200Kb to reduce file storage needs.
3. **Full Frame Image Capture:** DTS collects using full frame digital capture process (non-interlace) using full frame progressive scan cameras.
4. **Camera Setup:** The camera optics are routinely adjusted during collection to accommodate changing light (sunny to cloudy day) to optimize data collection efficiency.
5. **Multiple Viewing Angles and Camera Mounting:** There will be a minimum of three views during the right-of-way image collection. One view will be in front of the vehicle showing the

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driver's perspective, the second view will be positioned in the center of the van and the third view will be between 25 and 50 degrees to the right of the driving direction. DTS can also incorporate an additional view with a rear-facing camera providing for at least four views during the right-of-way image collection.

6. **Image Header:** All images will have a user-defined header, which includes date, route name and direction of travel.
7. **Camera Synchronization Requirements:** A DTS MAC vehicle captures all images at the same time with integrated results. Images can be integrated to any system as well as for development of the videologger.
8. **Digital Camera Frame Rate:** The camera systems can be configured to collect at a specific distance – triggered by the DMI.
9. **GPS/INS/DMI Requirements:** The DTS mobile mapping system produces a continuous GPS baseline utilizing a geodetic GPS receiver, an inertial measurement unit and an accurate distance measuring instrument (DMI).
10. **GPS Receiver:** The MAC vehicles use a 12-channel dual frequency L1 and L2, low noise and DGPS ready receivers. The DGPS solutions incorporate a base station or real-time differential link to both the US and GLONASS satellite constellations.
11. **Road Attributes:** All road networks will be segmented and tagged with centerline number and road names.
12. **Accuracy:** The horizontal position of any visible feature between up to 130 feet (40 meters) in front of the cameras will be one-meter root mean squared (RMS) or less (latitude/longitude).
13. **Shadows:** All cameras contain high dynamic range, so EarthShaper™ can be used to brighten an image to view features for extraction. An example appears below.



Cracking-Pavement Images

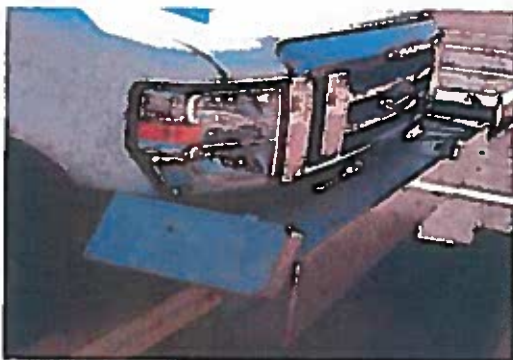
DTS' MAC vehicles utilize the INO-manufactured Laser Road Imaging System (LRIS), a downward-facing, progressive line-scan cameras (2 cameras), providing high-resolution panchromatic images of the pavement surface to clearly detect and quantify distresses. This system uses two angled laser illumination to create a shadowless, 1mm-pixel resolution image full of contrast. Our MAC LRIS system meets the requirement of AASHTO R 55-10 "Standard Practice for Quantifying Cracks in Asphalt Pavement Surface".

International Roughness Index (IRI) .Rutting and Faulting

DTS utilizes a surface profiling system manufactured by International Cybernetics Corporation (ICC) for evaluating the smoothness of pavement. The profiler uses infrared lasers and precision accelerometers to obtain accurate and precise profile measurements at speeds up to 65 mph. DTS MAC technicians have been trained and certified by ICC for the operation of road profilers and imaging systems. The proposed MAC vehicles for this project have been certified by ICC. Furthermore, DTS has received independent inertial profiler certification from the Texas A&M Transportation Institute in 2015, 2016 and 2017. DTS is scheduled for certification again in the summer of 2018. This independent accuracy verification ensures the City of Cheyenne the precision of the data being collected by the MAC vehicles.

DTS' MAC vehicles utilize International Cybernetics 5-laser profilers. A profiler uses three 16KHZ Selcom SLS 5200/300 point lasers and two 32KHZ Selcom SLS6375/750 angle lasers, providing a five-laser footprint with two single points in each wheel path, one single point center of profiler width and two angled lasers at 55 degrees (one left and one right). Wheel path spacing of lasers is 71 inches center to center of lasers.

The Selcom SLS 5200/300 profile lasers (wheel paths and center) have an accuracy of 0.0002 inches and they have a resolution of 0.002 inches. The Selcom SLS 6375/750 (angled) profiler lasers have an accuracy of 0.00037 inches with a resolution of 0.0037 inches.



DTS will use the Road Profiler Report System (WINRP) to process all collected IRI, rut and faulting data. DTS collects IRI consistent with AASHTO R 43-07. DTS collects the longitudinal profile measurements in accordance with ASTM E950 as a basis for estimating IRI. Roughness is reported for HPMS in IRI units of in/mi. Profiler equipment testing includes performing a bounce test and block test. The bounce test involves positioning the MAC collection vehicle on a flat smooth surface with calibrated flat metal plates under each laser. Then, a simulated DMI pulse is sent through the system. A DTS operator will bounce on the front bumper to displace the profiler from 1 inch to 2 inches. Visual assessments and actual test data is recorded from the output profile data. Visually, the laser heights should change when bouncing occurs. The recorded static portion should yield an IRI of less than 3 inches/mile before and after the 'bounce'. The bounce portion should yield an IRI of less than 8 inches/mile. If these two conditions are met, the profiler is operating optimally.

The block test is used to check laser height values. A calibrated straight metal beam is put beneath the entire profiler as a base point. Then, each laser is tested using 0.25, 0.5, 1 and 2 inch smooth machined blocks. The test is conducted using International Cybernetics WinPro application. The application will report a pass/fail grade on each block given the tolerance of 0.001 inches. If a laser does not pass a test, it will be retested and if fails again it will be re-calibrated and inspected for damage. IRI, rutting and faulting data will be collected to meet the requirements of AASHTO R 43-07. One method DTS has found to minimize IRI errors is to collect normally heavily congested areas on times where congestion is not an issue such as collection on weekends. DTS also uses flexible testing speed which allow collection of accurate longitudinal profile and roughness data at various speeds without loss of accuracy thus providing representative data even in congested areas while maintaining a bias of 5%. Lastly, to ensure accuracy DTS avoids sudden acceleration or deceleration.

Pavement Condition Rating and Evaluation

Condition Assessment

Pavement Condition Index (PCI) will be calculated per the ASTM D6433-11 Standard Practice for Roads and Parking Lots Pavement Condition Index Surveys for every sample identified in the City of Cheyenne's database. When necessary, inspectors will add "additional samples" to ensure an accurate representation of the Pavement Condition Index (PCI) for every section inspected. Each distress logged will include its respective severity level and quantity to provide the program with the information necessary to objectively and consistently compute the PCI and provide the City with a clear and specific picture of how each section is deteriorating. DTS is also familiar with numerous reporting methods, including mapping of all distresses in areas where pavement is evaluated and sample sections based on 95% percentile confidence. DTS will be available to discuss and plan for variable method of distress evaluation to meet the needs of the City of Cheyenne. Since we collect roadway and pavement images on 100% of the roadways driven, the method for distress evaluation can be modified based upon the City's desires for sample evaluation after field data is collected.

DTS will work with the City to determine the best treatment solution for each distress type and severity level, and will collect surface distress for flexible/composite pavement and for concrete pavement using the same methodology as MicroPAVER.

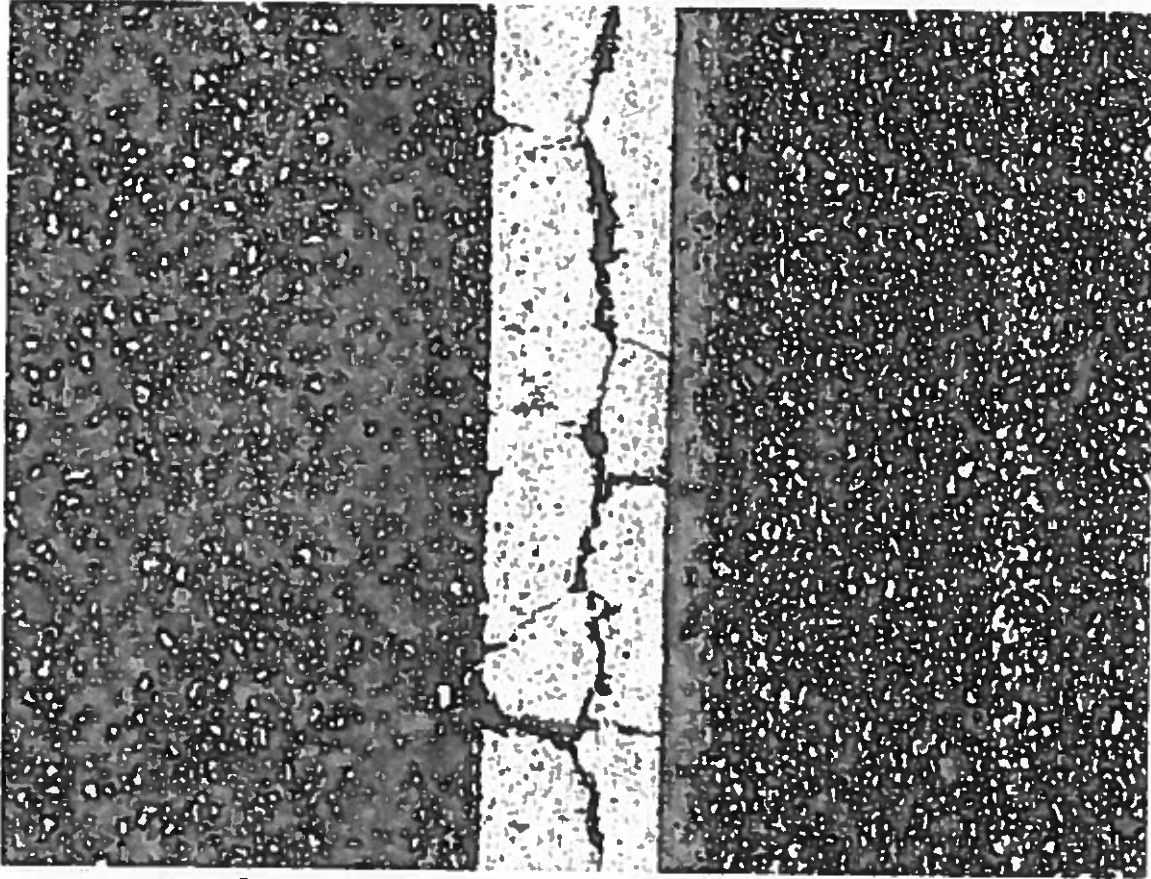
Pavement Surface Imaging Rating

DTS MAC LRIS vehicle pavement imaging sensors are oriented from nadir (straight-down) to achieve the best perspective, laser-illuminated to ensure uniform image contrast and GIS-integrated to provide geospatial distress vectors (points, lines and polygons) that can be loaded and verified using GIS.

- DTS will utilize a downward-facing, progressive line scan camera that provides high-resolution images (1mm pixel, 4,000 pixels wide and ~12 feet width) of the pavement surface to clearly detect and quantify distresses.
- pavement surface imaging (JPEG format) will span, at a minimum, the data collection lane from left lane stripe to right lane stripe, and will provide 100% continuous pavement coverage
- image resolution will be such that all visual cracking distresses can be accurately identified and quantified
- images will have a minimum horizontal resolution of 4,000 pixels or better

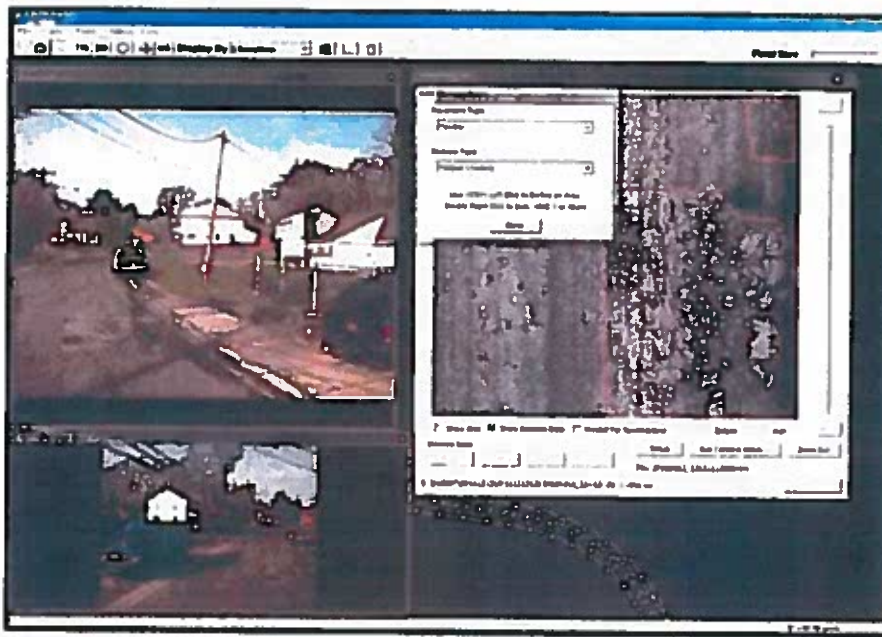
DTS Technical Proposal to City of Cheyenne RFP 20 – 18 – Pavement Condition Survey Project

- Images will be synchronized with DTS centerline file
- DTS will collect longitudinal profile and roughness data (IRI) to provide a ride condition index for each segment



Pavement Image Captured with the 4K Laser Road Imaging System

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DTS will use our asset extraction software - EarthShaper™ - to extract pavement data for the project. DTS begins the pavement distress rating process concurrent with image collection. Once pavement image and distress mapping processing are complete for each collection day, DTS' experienced pavement inspectors review each street

segment's images for a complete and thorough evaluation of the existing pavement condition per the ASTM D6433-11 concrete and asphalt pavement distress rating process for calculating the PCI value including the following distresses: alligator cracking, longitudinal cracking, transverse cracking, raveling, flushing and patching. DTS team of pavement inspectors follow the ASTM standards, to ensure that all samples fall within the allowable size and range and will be determined to accurately represent the pavement section. The EarthShaper software gives DTS a distinct advantage over competitive software packages because it removes the "black box" portion of pavement rating.

All distress vectors can be viewed and edited through this workflow, resulting in a more accurate product for the City of Cheyenne. It also adds another element to the QA/QC process, allowing for more accurate data review. The EarthShaper software is capable of accommodating any asset/attribute combination through a configurable data model. This model can be set up specifically for the the City of Cheyenne and is 100% Esri-integrated to streamline the data capture process into a GIS.

One capability unique to DTS is that we deliver an ArcGIS shapefile containing each longitudinal, transverse and alligator distress identified and categorized by severity. Distress mapping provides an additional 'point in time' reference for the pavement condition. GIS mapping of pavement distresses is a unique tool for guiding decisions while evaluating funding needs for the City of Cheyenne.

DTS is very experienced with collecting and reporting ASTM E950 and ASTM D6433 data. IRI (International Roughness Index) will be collected using equipment that meets ASTM standards. DTS utilizes a surface profiling system manufactured by International Cybernetics Corporation (ICC) for evaluating the smoothness of pavement. The profiler uses infrared lasers and precision accelerometers to obtain accurate and precise profile measurements at speeds up to 65 mph.

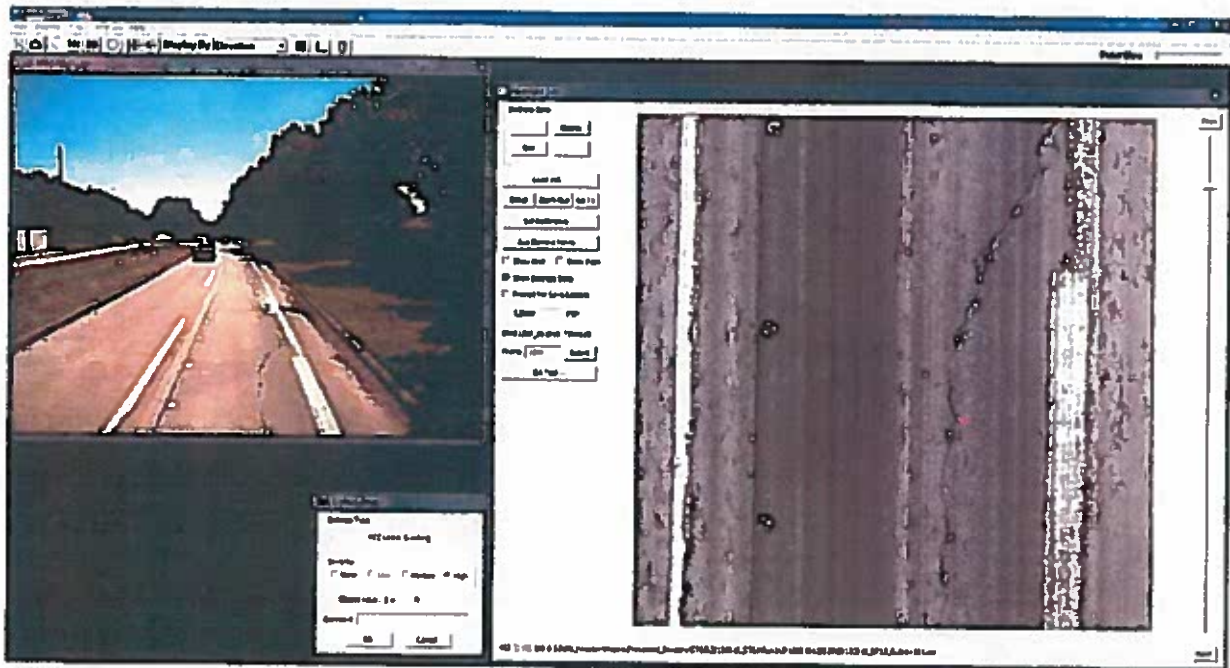
DTS Technical Proposal to City of Cheyenne RFP 20 – 18 – Pavement Condition Survey Project

Surface Type Confirmation

Asphalt Distress Rating



Concrete Distress Rating



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Verification of Section Width

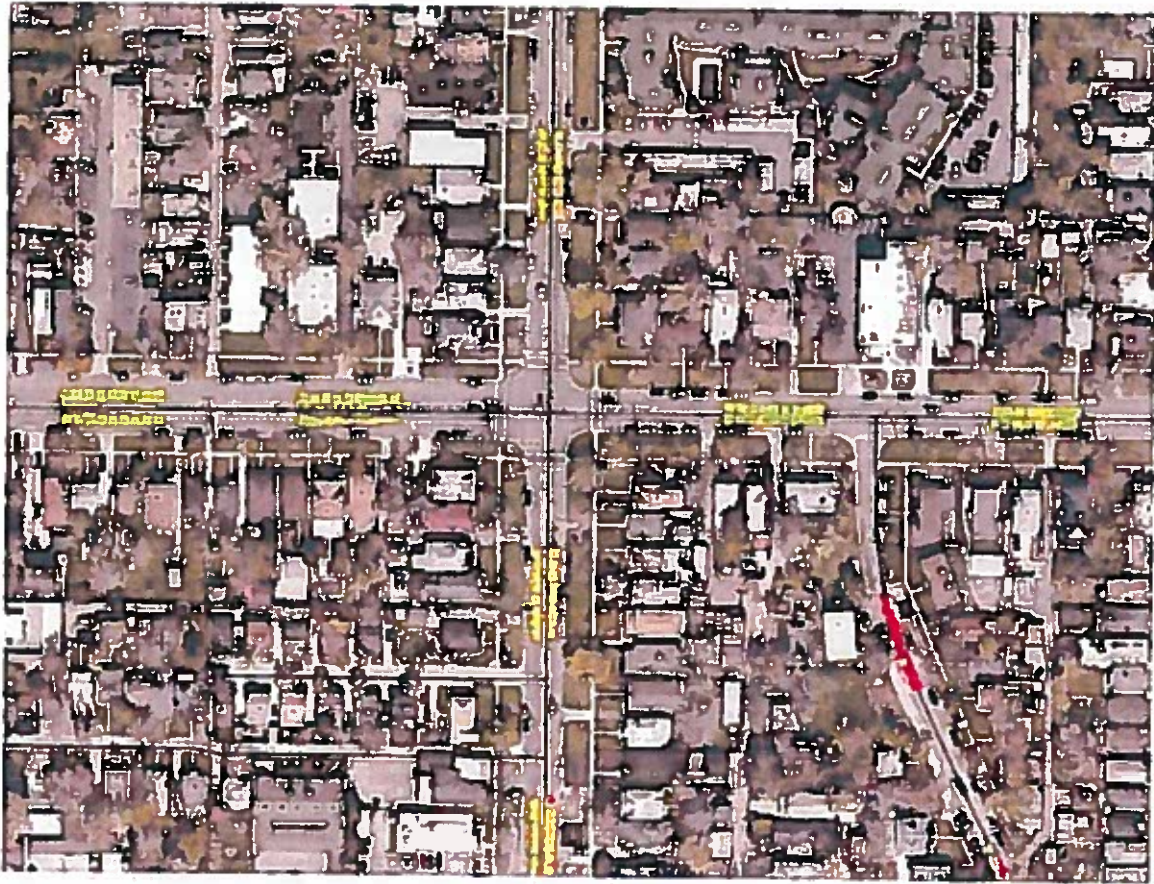
Pavement Length and Width Calculation

DTS will determine each road segment's length and width. The length will be calculated based on the centerline in the GIS layer. The width data will be obtained from the City GIS record. If this record is not available in the City GIS, DTS will work with the City to, at minimal cost, perform width measurement using the captured photos and aerial photography to be provided by the City. Width will be measured at locations where a significant change in width occurs. To keep the project cost low, minor width changes such as turning lanes, widened lanes at intersections, etc. will not be captured in this run of the project.

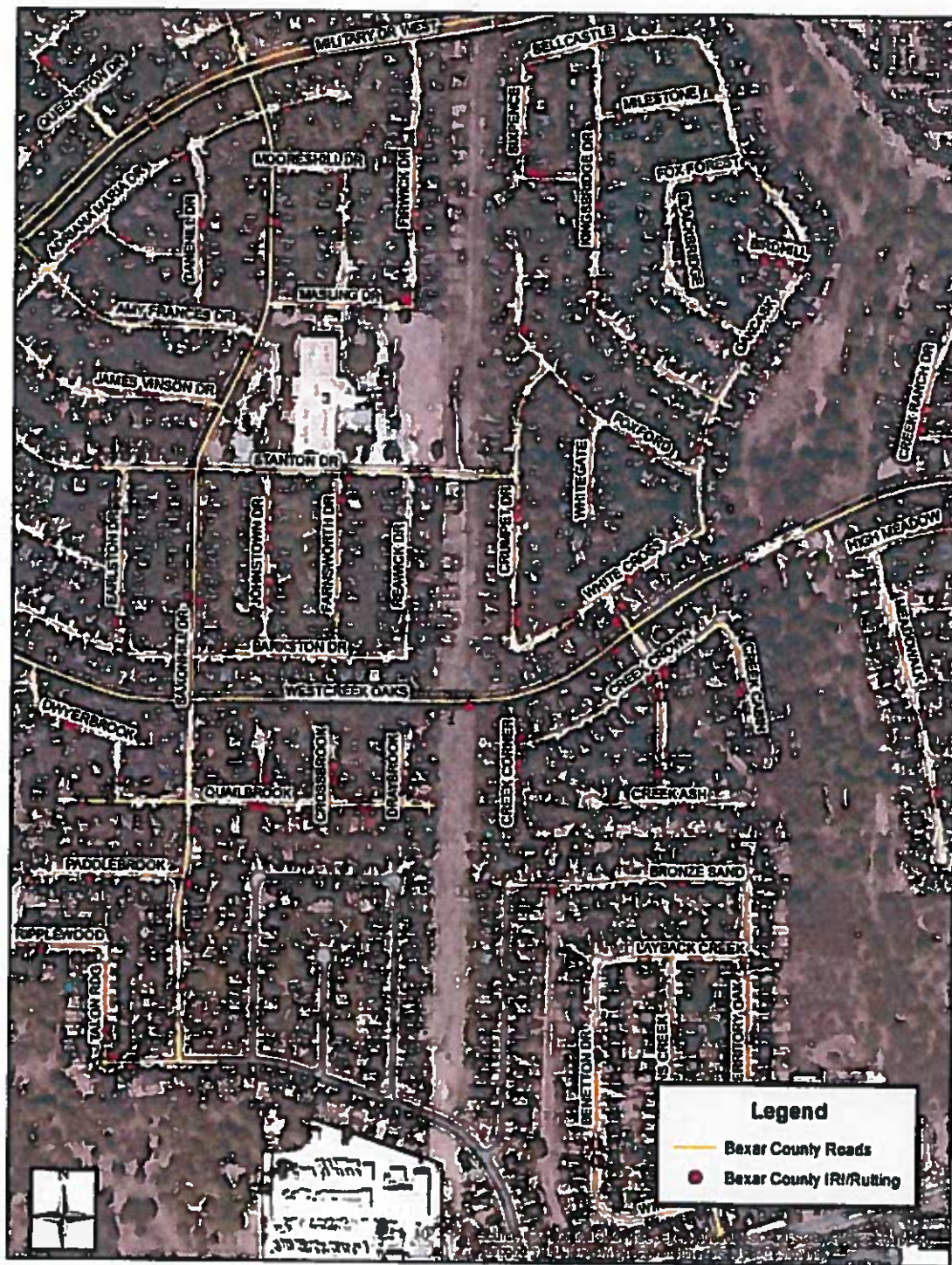
Using the MAC right-of-way imagery, DTS will measure the width of pavement for each street segment.



Example of Pavement Width Shown in Imagery (EarthShaper Software)



Pavement Distresses Mapped in GIS



IRI Displayed in Maps

MicroPAVER

Data Entry and Update the PAVER Pavement Database

DTS will facilitate the inspection (interpretation) of the collected data, and the uploading of quality checked inspection data in the City's current PAVER database. This will be performed by trained inspectors that are proficient with PAVER and strictly follow ASTM standard guidelines. The team will accurately identify distress type, severity level and quantity of each distress found in every sample - for all sections established in the City's PAVER database. Once the data is uploaded, QC checks will be performed on the database to ensure there are no missing or bad data points with a QC check on every section's latest inspection record with corrections being made if necessary. The QC process will confirm the following: every section included in the project received an updated inspection; duplication of distress entries are found and corrected; questions about specific distress types and quantities are confirmed by inspectors and corrected when necessary and, if needed, the project manager or pavement engineer will revisit any section where inspection details need to be validated.

Create Pavement Management Scenarios

DTS will complete this task in three phases:

1. DTS will create Prediction Models for the City based on the updated inspection data. These models will serve to increase the level of accuracy with which the work plan predicts future conditions. Families – for the purposes of generating functional models – will be modified or created based on surface type and section rank designations and the strength of each model will be cross checked statistically.
2. Updated cost tables will be created specifically for the City that reflect both current M&R practices and associated costs. DTS will consult with key City personnel to ensure accurate cost data is being used during work plan generation.
3. DTS will use inputs from the City's PAVER database and key personnel to generate several work plan scenarios that will demonstrate impact on condition and penalty cost of deferred maintenance. Comparison of the scenarios will give the City the ability to quantify the return on investment for various strategies and provide justification for proposed budgets. Scenarios will include 5, 10 and 20 year views and will incorporate the consequence of different budgets and the cost of achieving specific goals – including the cost of maintaining network average PCI of 80 (if determined to be feasible), an unlimited budget option, and City provided budget options.

The PAVER software will be used for the City's Pavement Management Program (PMP). The PAVER software offers benefits to the City inclusive of the following:

- Low cost solution
- Continual software development and updates supported by FHWA, APWA and various other agencies
- Simple geospatial integration with other asset management geospatial tools
- Non-proprietary

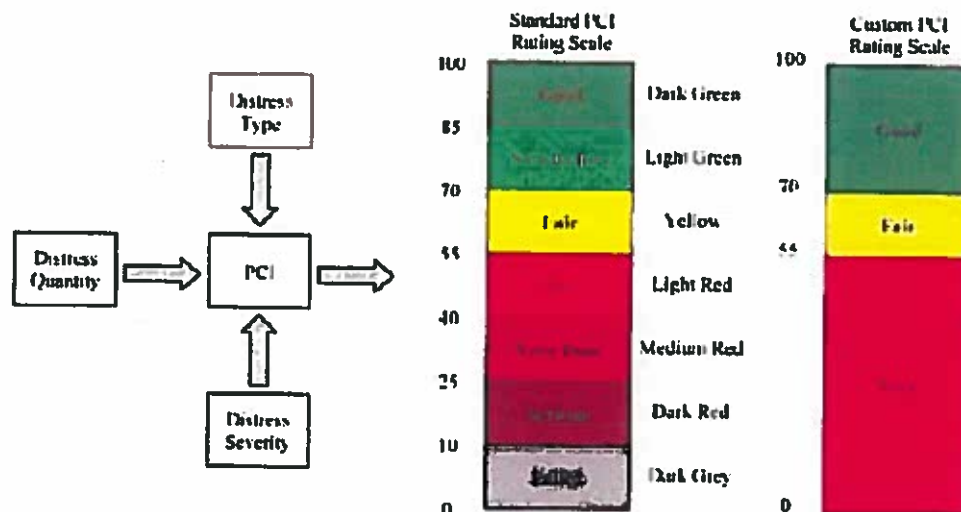
The PAVER software allows the user the ability to combine multiple segments or blocks into sections through the creation of a virtual database. This is a powerful tool already used by the City to help make thoughtful and cost effective maintenance and rehabilitation (M&R) decisions through proximity



Scott McDonald, APM is an acknowledged technical expert in the use and support of U.S. Army Corps of Engineers (USACE)-developed Sustainment Management Systems (SMS) products, including BUILDER and PAVER. He has provided PAVER pavement management system implementation assistance and infrastructure asset management support and training for a variety of federal, state, municipal and higher education clients, including the US Air Force (USAF); The FBI Academy; the City of Rio Rancho, NM; San Diego County, CA and NASA. Mr. McDonald also provides software support and training to the American Public Works Association's (APWA) PAVER user community and has implemented the PAVER pavement management system for hundreds of thousands of miles of airfield and road/parking lot network pavement throughout the U.S. and abroad.

DTS will utilize GIS data provided by the City to build the pavement network within PAVER. The network will be comprised of branches and sections as described in PAVER terminology. A 'Network' is the whole area of pavement under investigation which in this case is the City. A 'Branch,' is an individual road within the network, for example E. Cary Street. A 'Section' is a homogeneous segment within the branch that was constructed at one time, has the same traffic and is in a consistent condition throughout. To coordinate with the City's GIS, this will be synonymous with the Facility ID on the feature class layer in GIS to allow for data translation between GIS and PAVER. A section is typically one block long. As will be discussed within the pavement condition survey task, a sample unit (as defined by ASTM) will be located within the section within which pavement data will be collected.

The integration of the GIS database into PAVER will be done simply through PAVER import tools. The process is quick and seamless and has been done extensively by the DTS team over the last several years. The PAVER database is primarily used to collect pavement distress data, however it can handle additional non-pavement inventory and condition data through either predefined or 'user-defined indices'. PAVER, however, aside from pavement width, does not utilize these inventory items within any of the analysis modules. These data should be layered with pavement condition in GIS in order to make engineering judgments of repair needs.



PCI Ranges can be Customized for Reporting and Analysis

The DTS team has vast experience on training with pavement management systems and will provide the City training on PAVER. This training will cover the basic PAVER functions to the advanced reporting solutions within PAVER. The training will be performed on the City's actual data that is collected during the project. Highlights of the course include fundamentals of pavement management, Pavement Condition Index procedure, pavement conditions prediction, project-level management, network-level management and the PAVER computer system. Our typical class outline for PAVER training is listed below:

- a. Overview of PAVER - Selection tools and dropdowns
- b. Inventory - Network, surface type, user defined fields

PAVER inventory management training is based on a hierarchical structure composed of networks, branches and sections, with the section being the smallest managed unit. This structure allows staff to easily organize their inventory while providing numerous fields and levels for storing pavement data.

These are some of the other features included in Inventory:

- **User-defined Fields:** In addition to the standard inventory information, users can define their own fields to meet their management requirements.
- **Virtual Inventory:** Allows the user to create virtual copies of the existing inventory and group sections for easy presentation.
- **Surface Change:** Automatically calculates and updates pavement surface based on work history information.
- **Edit Historical Inventory:** Easily edit historical inventory values associated with previous inspections.

c. **Work Planning.** Trains users on the ins and outs of the Work Planner; this tool is used for planning, scheduling, budgeting and analyzing alternative pavement maintenance and repair (M&R) activities. Users will also learn the two ways to analyze budgets scenarios. First, they will learn how to determine the consequence of a selected budget on pavement condition and the resulting backlog of Major M&R. In addition to a single budget scenario the budget split feature is also taught. The budget split feature will allow City staff to split a budget based on different M&R work types. Second, they learn how to determine the budget requirements to meet specific management objectives, e.g. backlog elimination or PCI goal. This will enable staff to develop a variety of funding scenarios to support their decisions.

d. **PCI Field Inspection/Data Entry**

The staff will learn ways to assess pavement condition using PAVER PCI. The PCI measures pavement condition on a scale from 0 to 100. ASTM has adopted the PCI as standard practice for roads (D-6433-11). This section provides users the ability to customize the PCI condition rating categories. In addition to the PCI, managers learn to use and create other condition indices, including those based on PCI distresses. A new interface has been added for easily importing inspection data from automated vehicle collection sources.

e. **Prediction Modeling**

The Prediction Modeling training section in PAVER will help the users identify and group pavements of similar construction that are subjected to similar traffic, weather and other factors affecting pavement

performance. The pavement condition historical data is used to build a model that can accurately predict the future performance of a group of pavements with similar attributes.

f. Condition Analysis

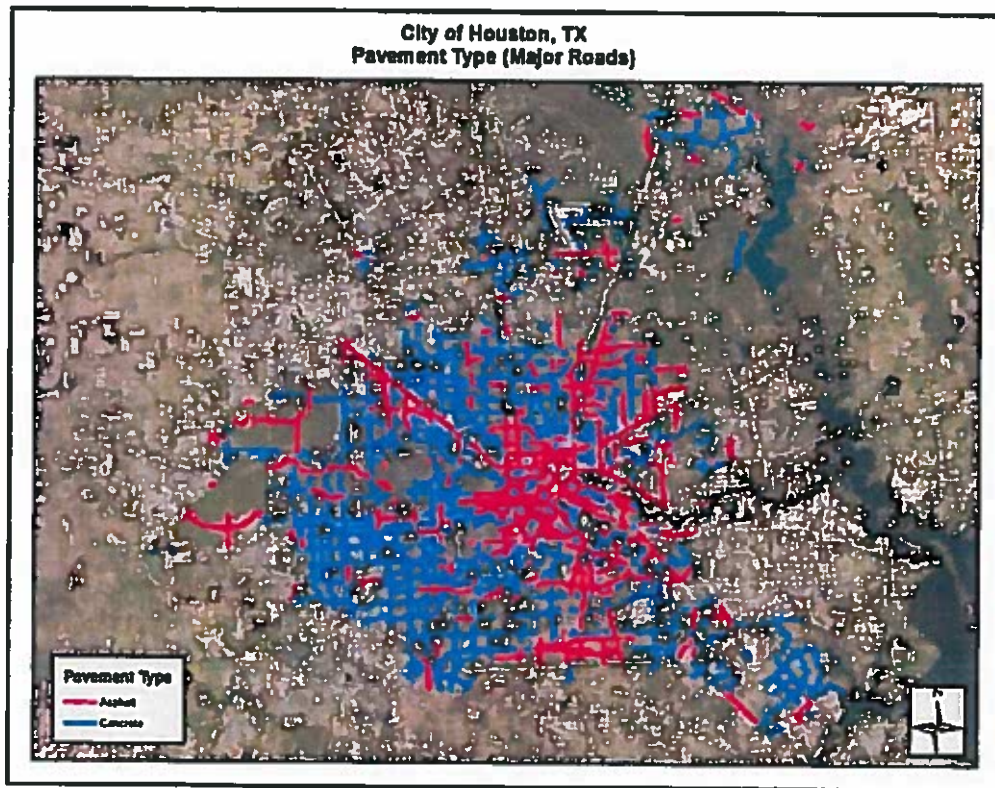
The Condition Analysis feature training teaches users to view the condition of the entire pavement network or any specified subset of the network. In this section, personnel will learn to use feature reports of past conditions based on prior interpolated values between previous inspections. These reports project conditions based on prediction models. The conditions can be viewed on GIS maps in addition to tables and graphs.

Pavement inventory, work activities and inspections information are exported to PAVER to calculate the PCI (Pavement Condition Index), perform pavement analysis and utilize past pavement work histories to predict future pavement conditions. PAVER is also of significant value in planning pavement maintenance and rehabilitation (M&R) because it can set priorities of work to be done according to pavement condition and/or limit the work to a budgeted amount and available resources.

Reports, Presentation and Training

At the conclusion of the data collection effort, DTS will prepare and present a final report to City staff. Additionally, DTS can assist the City with presentations to the City Council. Additionally, training will be provided to staff appointed by the City on the use of the PAVER system. With 18 years of experience providing training and implementation assistance to municipalities around the country, DTS will provide training materials to students and share experiences, current technologies and best practices as a component of the training. The goal of staff training will be to ensure that the City has the knowledge to make best use of the data provided by its PAVER system. This will give the City the best opportunity to reduce the penalty cost of deferred maintenance and increase return on their pavement maintenance investments. DTS will submit a pavement summary report as well as the PAVER e70 file to the City.

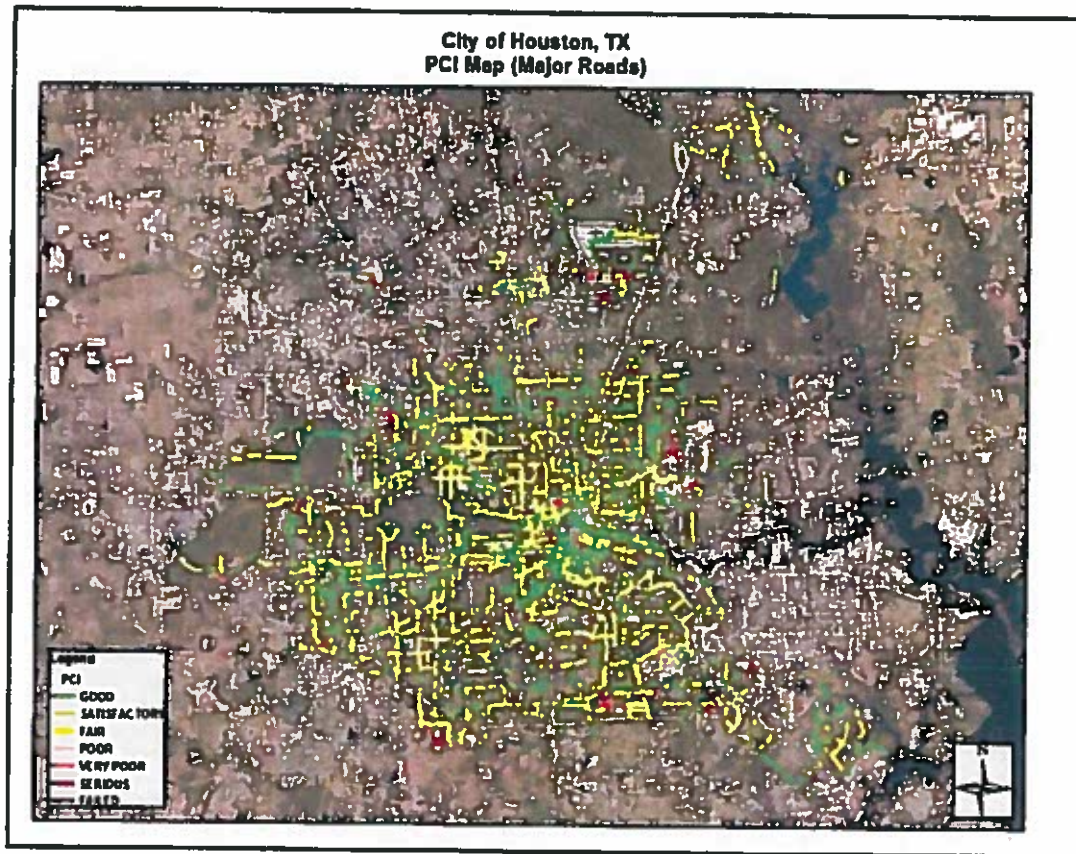
Below are captions from a Sample Pavement Report DTS prepared for City of Houston, Texas:



Review of the City's PCI values shows a general breakdown of street segments by condition to be:

Pavement Condition Index (PCI) Range	Condition Description	Percent of Network	Legend
86-100	Good	45.52%	
71-85	Satisfactory	34.35%	
56-70	Fail	12.93%	
41-55	Poor	4.84%	
26-40	Very Poor	1.85%	
11-25	Serious	0.44%	
0-10	Failed	0.07%	
Total		100%	

City of Houston (Major Roads) PCI Ranges by Percent



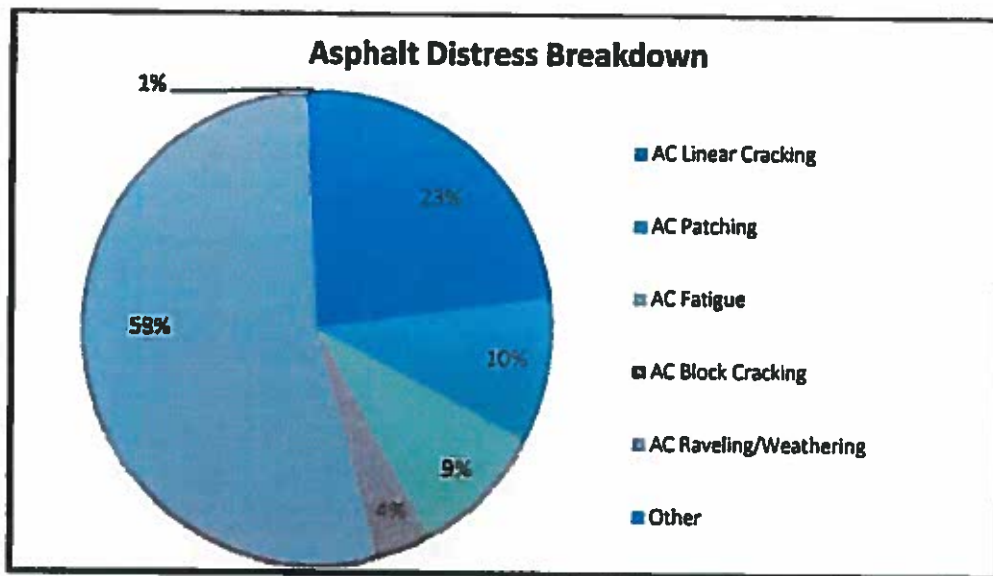
City of Houston (Major Roads) Network PCI Map

Breakdown of IRI by Pavement Type

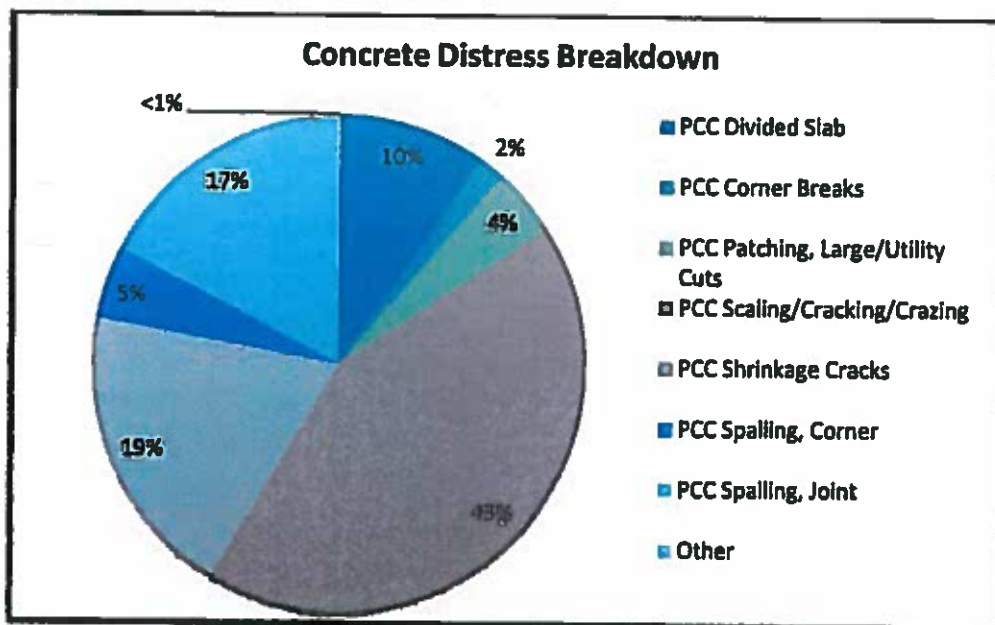
Pavement Type	AVG IRI
Asphalt	270.41
Concrete	303.51
Major Road Network	293.95

City of Houston Major Roads Average IRI

The breakdown of the types of distresses for both concrete and asphalt can be seen below.



City of Houston Asphalt Distress Breakdown



City of Houston Concrete Distress Breakdown

Concrete Distress Breakdown by Severity			
TYPE	SEVERITY		
	LOW	MEDIUM	HIGH
PCC Bleeding & Pumping	0.053%	0.026%	0%
PCC Corner Breaks	1.060%	0.461%	0.089%
PCC D Cracking	0.009%	0.007%	0.001%
PCC Divided Slab	7.079%	2.205%	0.355%
PCC Lane/Shoulder Dropoff	0.031%	0.021%	0.020%
PCC Patching, Large/Utility Cuts	3.433%	1.030%	0.131%
PCC Punchouts	0.031%	0.002%	0.002%
PCC Scaling/Cracking/Crazing	36.627%	6.086%	0.142%
PCC Shrinkage Cracks	19.017%	0.144%	0.010%
PCC Spalling, Corner	2.706%	1.647%	0.312%
PCC Spalling, Joint	10.927%	5.435%	0.901%

City of Houston Concrete Distress Breakdown

Asphalt Distress Breakdown by Severity			
TYPE	SEVERITY		
	LOW	MEDIUM	HIGH
AC Edge Cracking	0.010%	0.194%	0.060%
AC Lane/Shoulder Dropoff	0.008%	0.038%	0.019%
AC Linear Cracking	0.199%	18.915%	3.785%
AC Pothole	0.001%	0.002%	0.002%
AC Bleeding & Pumping	0.003%	0.212%	0.005%
AC Block Cracking	0.155%	1.904%	1.726%
AC Fatigue	0.692%	2.985%	5.572%
AC Patching	0.173%	6.872%	2.895%
AC Raveling/Weathering	0.061%	53.043%	0.438%
AC Slippage Cracking	0.028%	0.002%	0%

City of Houston Asphalt Distress Breakdown

Below is a summary of the Roadway Network PCI based on surface type:

Pavement Type	City of Houston, TX Major Roadway Network						
	Good (86-100)	Satisfactory (71-85)	Fail (56-70)	Poor (41-55)	Very Poor (26-40)	Serious (11-25)	Failed (0-10)
Asphalt	10.46%	9.23%	4.90%	2.74%	1.58%	0.43%	0.06%
Concrete	37.12%	24.63%	6.83%	1.60%	0.39%	0.02%	0%
All	47.58%	33.86%	11.73%	4.34%	1.97%	0.45%	0.06%

City of Houston Major Roadway Network by Pavement Type with PCI Percentile

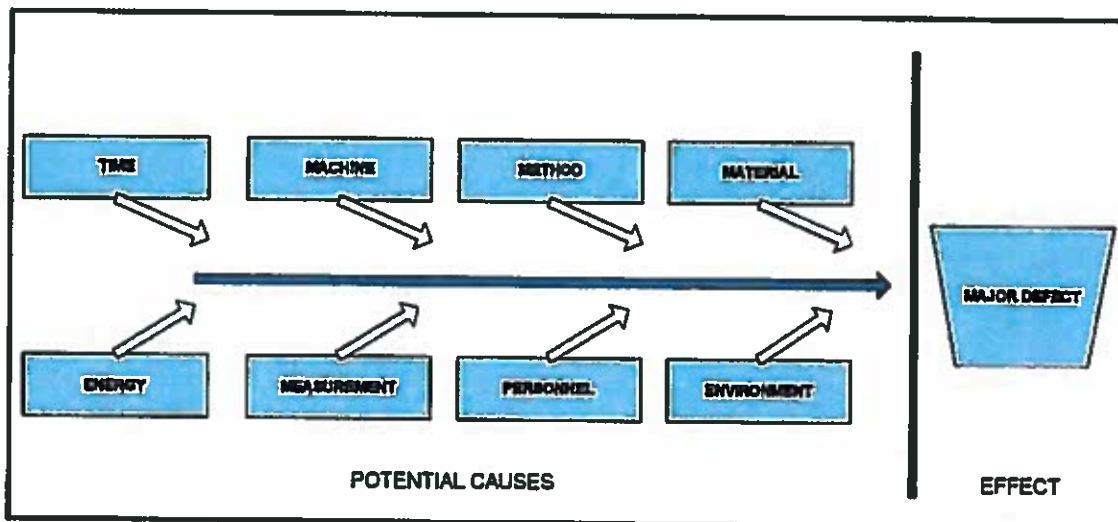
Quality Control/Quality Assurance Plan

DTS will only conduct pavement survey work on dry pavement and in lighting conditions that assure accurate crack detection. DTS will collect imagery during daylight hours only, with no rain, smoke, fog or snow visibility obstructions. DTS uses automatic shutter adjustments to allow for on-the-fly image brightening or darkening to avoid poor image quality from sunlight, overcast or shadows. Any road segment that exhibits low image quality due to lighting will be recollected.

Quality Control involves the process of monitoring and recording the results of activities to assess the performance of the project team, the quality of the deliverables and the process of recommending and implementing any necessary changes. The DTS team follows the Project Management Institute (PMI) standards for Quality Control, and our processes are overseen by PMI-certified Project Management Professionals (PMPs). Our QA methodology involves routine audits of project deliverables to ensure they comply with quality requirements. In a broad context, our process includes the collection of quality metrics throughout the execution of the project. These metrics include performance measurements such as:

- planned vs. actual performance
- planned vs. actual schedule performance
- planned vs. actual cost performance

When issues are encountered, we utilize a variety of diagnostic and corrective techniques. These include cause and effect diagrams, Ishikawa diagrams and fishbone diagrams. Generally similar tools, these diagrams help us determine the possible root cause for any defects, delays or deficiencies. The benefit to our process – and ultimately the City – is the learning gained from these analyses. More specifically, when we encounter an issue, we do not just address the issue, we address the root cause of the issue, ensuring to the maximum extent feasible, that the issue is no longer a factor. Additionally, through our own Organization Process Assets, we maintain a record of all our lessons learned so that the City also benefits from our past experiences.



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The QA team is committed to providing City with quality products and professional services. No deliverable shall be submitted without having first undergone the QA/QC processes outlined and described herein. Please note, while this is *our team's formal QA/QA procedure*, it is not the only acceptable procedure. An alternative procedure may be more appropriate in some cases, or the City may require specific QA/QC procedures.

DTS employs an originator/checker process for processed and compiled data where the project engineer (originator) provides a qualified project QC checker to review and check processed and compiled data. The QC checker reviews calibration/correlation data and imagery for data anomalies and inconsistent pavement data, false positives (incorrect distress) pavement distress vectors and quality of post processed MAC imagery. The project manager, engineer and QC checker determine if any errors found require a run to be recollected and processed or reprocessed and reviewed again.

After the pavement condition has been evaluated, a team led by Pavement Engineer G. Scot Gordon, PE will review with City staff the pavement analysis quality. DTS provides walk-out maps that display pavement distress data for field confirmation and acceptance.

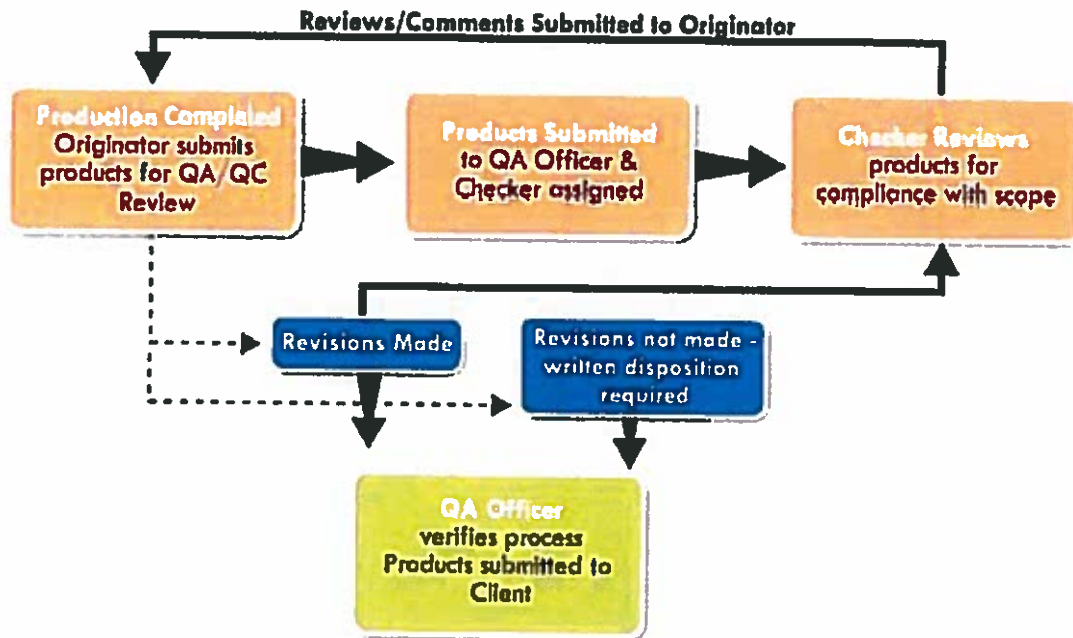
Team Roles and Responsibilities

- **QA/QC Officer (Technical Manager)** – the officer in charge of assuring the QA/QC plan is followed for each project submittal
- **Originator (Project Manager)** – the individual who assures all documents are complete and ready for the Checker's critique
- **Checker (QC reviewer, not associated with project details)** – one or more qualified individuals responsible for verifying that project document/plans are complete and meet the requirements of the project scope of services
- **QC Facilitator (optional)** – a person who assists the Originator with the QA/QC process by responding to Checker comments and/or making necessary changes to the contract document/plans as a result of comments made by the Checker.

Standard Operating QA/QC Procedure

1. The Originator assembles all required project documents. The Originator verifies all documents are complete and ready for a QA/QC review.
2. The documents are then provided to the QA/QC Officer for assignment to a Checker.
3. The documents are reviewed by the Checker for compliance with the project scope of services. Any comment made by the Checker is properly documented.
4. Once the review is complete, the document is returned to the Originator for response to comments and/or corrections.
5. The Originator reviews the comments and then meets with the Checker to ensure the comments are understood. The Originator will either accept the Checker's comments and make the necessary changes or will reject the comment and leave the document as originally submitted for QA/QC. All rejected review comments require a written response on disposition.
6. This cycle will be repeated in as many iterations as necessary until the Checker is satisfied there are no additional changes and revisions required.

7. The Originator assembles the final corrected, reviewed document/plan, along with the previous marked review document/plan, and all documents are then returned to the QA/QC Officer.
8. The QA/QC Officer verifies the process has been followed and there are no outstanding comments prior to submittal to the Client.
9. Upon final submittal, the QA/QC documentation, including marked up/edited drafts, comment dispositions and forms will be archived and kept with the project files.



Specific to this project, the DTS team understands the level of QA/QC required of this delivery to be quite high – a standard to which our team's firms routinely holds its deliverables. The DTS team QA process includes the full complement of the Data Quality Management Cycle, including:

- **Define data quality** – Identify the acceptable levels of resolution, accuracy and repeatability.
- **Plan and implement QC** – Develop and implement a set of procedures to produce, check and ensure data of acceptable quality.
- **Perform acceptance tests and evaluate results** – Perform tests to compare delivered data to acceptability metrics.
- **Take corrective action** – Take steps to re-collect or reprocess data as needed to achieve data acceptance standards. This is the step where the DTS team remediates any deficiencies identified and provides a prompt timeline for returning to acceptable quality standards. Upon identification, corrective measures are taken immediately to resolve the issues. Because of the previous experience and extensive QA/QC process issues are identified quickly and have a swift resolution. For instance, if image quality becomes an issue the MAC Operator immediately recognizes the equipment issue creating inadequate image quality, ceases collection and determines the proper remedy for the poor image quality.

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- **Report on data quality** – Document the data quality standards, protocols, equipment, personnel, collection and processing methods, QC, acceptance tests and results.
- **Improve the process** – Use the knowledge and experienced gained to modify processes as needed to improve data quality.

Equipment Calibration and Method Acceptance

Testing equipment is calibrated and testing methods and analysis are accepted prior to data collection and checked periodically thereafter to verify that the equipment is functioning according to expectations and that the collection and analysis methods are being followed.

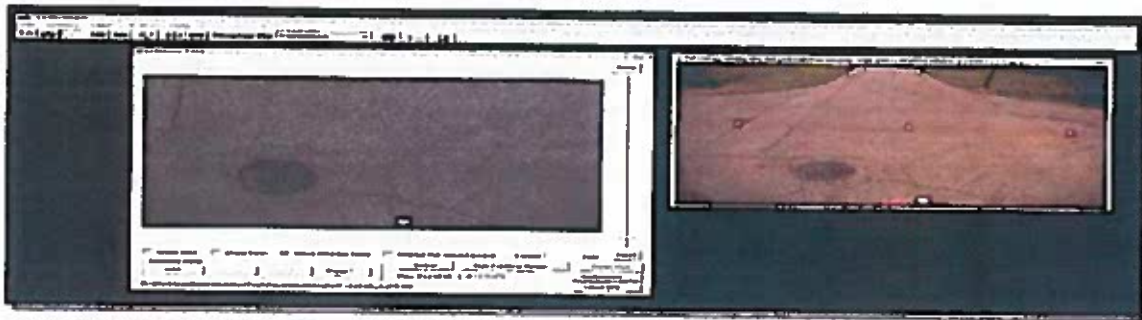
DTS' MAC LRIS system is certified annually by International Cybernetic Corporation for laser repeatability and accuracy. DTS MAC operators have also received LRIS collection training by International Cybernetic Corporation staff. Additionally, DTS has rigorously tested equipment that is independently certified by the Texas A&M Transportation Institute (TTI). The TTI certifications were obtained in 2015, 2016 and 2017 and are again scheduled for certification in 2018.

The equipment used for automated network-level data collection consists of complex sensors, computers, GPS units and video systems installed on the mobile asset collection vehicle (MAC). The components are configured according to agency requirements and calibrated as specified by the manufacturer. Calibration of the laser profiling system includes laser sensor checks and block tests to ensure the accuracy of the height sensors, accelerometer calibration "bounce tests" to verify proper functioning of the height sensors and accelerometers and distance calibration to ensure accuracy of the Distance Measuring Instrument (DMI). Calibration of the DMI and some accelerometers occurs during field testing, and each is checked and recalibrated on a regular basis. The DMI calibration includes testing on a course of known length, and of sufficient length to minimize calibration errors. If a sensor is replaced, the block test is run to verify that the replacement sensor is performing accurately. Agencies typically conduct "bounce tests" and other equipment tests on a weekly basis.

The MAC technician will check each camera's exposure rate, image quality and GPS and IMU operation to ensure the MAC system is recording the image, GPS, DMI and IMU data and that the GPS location is within the stated project tolerance. Each collection day's calibration collection will be documented in the MAC collection log book. The MAC collection log book also contains information such as date, location, technician and driver name, any issue that developed during the collection day and DMI calibration runs. DTS maintains a Microsoft Access database of any collection or other project issues. All project team personnel including City personnel will have access to the database to log comments, check the status of issues and have one central repository to track issues and resolutions.

Control Site Testing

The MAC calibration/control site, which serves as a quality control for data collection, will consist of a minimum of 10 grid pavement points that are surveyed for a precise and accurate image daily calibration check. Each MAC vehicle will collect the calibration site every day of collection for the project duration as a MAC system QC item. This process ensures that each day's run can be calibrated based on the accuracy requirements (sub-meter) of the project. The MAC system is capable of collecting all right-of-way images and pavement data concurrently.



Calibration Site Checked Daily to Ensure the Accuracy of Collection



On-board Real-time Verification of Equipment and Collection Operating Correctly

During image collection, the MAC technician reviews the images collected on-screen as they are collected and any issue with image clarity requires the collection run to end and the image quality issue to be resolved. This provides real-time verification that the equipment is operating correctly. Once resolved, the collection run begins from the beginning for the road segment collected. The MAC technician also monitors GPS reception during collection. If GPS reception is lost measured using PDOP (positional dilution of precision), the MAC technician stops the collection and resolves the GPS reception issue. Collection

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begins again once the GPS reception issue is resolved. All issues resulting in the collection run being stopped will be recorded in the MAC collection log book along with the resolution.

Below are examples of algorithm processing and reporting process and policies to ensure the highest accuracy of the data being collected.

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In addition to calibrating the equipment, our QA/QC staff will work with the City to establish the list of data items that will be used in the measurement calculations. Onboard software generally analyzes the IRI, faulting and rut statistics from the raw longitudinal and transverse profiles.

Personnel Training

Field crews are trained to calibrate, operate and troubleshoot complex equipment; raters trained on the proper protocols and pass competency tests and data reduction personnel must learn how to process, compile, properly format, segment and check the data for errors.

Distress rating technicians (for either manual or semi-automated surveys) undergo extensive training before production rating begins. Training is carried out and evaluated, pavement condition raters are instructed in the agency's unique distress rating protocols. In addition, raters conduct and are evaluated on verification sites specific to the pavement condition data collection. Distress ratings performed by the survey crews are compared to those of the lead rater for accuracy and consistency. Cross-rater checks are

also performed to ensure all raters produce similar results. If any raters are found to be deficient, additional training may be required.

The training program for automated data collection crews is also extensive. Training typically includes calibrating the equipment, monitoring the data and video systems in real time and understanding the multitude of factors that can affect data quality during the collection process. There are also a series of startup checks (e.g., laser calibration, accelerometer “bounce test”, distance measuring instrument (DMI) that are performed daily before collection begins to ensure proper functioning of equipment and software. There are also checks of systems and data that are performed at the end of collection every day (e.g., data completeness, image and data quality) and other systems checks that are performed on a weekly or periodic basis (e.g., DMI and GPS). Finally, field technicians are trained to review data in the field to look for inconsistencies and to check error logs. This allows the crews to spot problems, take corrective actions and re-collect the segments before leaving the area. Field technicians keep a proper log of all checks to submit to the project manager as documentation.

Field crews have checklists of daily checks that must be done at the beginning of each day, including checking the computer systems, camera enclosures, tire pressure and lasers. In addition, the field crew should check video monitors periodically throughout the day to watch for any changes in video quality. The crew checks the onboard road sections and reference point database during collection to ensure the proper segments are being collected. Shapefiles, which are used to store geometric location and associated attribute information, are loaded into the computer to facilitate identification of collection segments. At the end of each day of data collection, field technicians back up data and review data files to look for irregularities. Log files keep track of all error messages. These, as well as samples of video, are reviewed for any issues that need corrective action. The field crew fills out a daily log of activities, including QC performed. Reports are uploaded and reviewed by QC personnel in the office.

Distress Rating Checks

The QC program for pavement condition data collection typically includes random sample audits, inter-rater reproducibility and data checks for accuracy and repeatability of the results. Random samples of the pavement condition data may be selected and checked by the lead rater or QC personnel. If the pavement condition ratings do not meet quality standards, corrective action is taken, such as retraining of raters or retesting.

Checks of distress ratings are a manual process in which samples of data are visually inspected for accuracy of the ratings. If production ratings were performed by windshield survey, the acceptance checks will usually be done by windshield or walking survey. If semi- or fully automated distress reduction methods were used, acceptance checks will be performed by viewing the pavement images. The data collection vehicle includes software programs that overlay the distress ratings directly on the pavement image to allow viewing of the image and ratings together. As the distress rating samples are checked, a log of all data checked and the results of the inspection are recorded. Since distress rating checks are very time-consuming, we will commonly begin sampling checks before data collection/rating is complete. Because of this overlap, there is opportunity to promptly re-collect or resurvey any sections with data that do not meet quality criteria.

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We will define the values for each collected data element. Context-based validation rules can be programmed into the rating software so that inconsistent or illogical data, based on the event context, cannot be entered. Dropdown menus can also be used to minimize the key strokes and simultaneously improve data accuracy.

Inter-rater reliability (also called cross-rater) means that two or more competent raters using the same protocols on the same sample sections should get the same results. Repeat test reliability (also called intra-rater) checks are performed by giving a single rater a sample of pavement to re-rate later in the data collection phase and comparing the ratings with the previous ones. If similar results are not produced on any of these checks, further training and rerating of batches should be performed. A logic check is another type of check performed on distress. A logic check looks for consistency between the types of distresses rated and pavement type (e.g., asphalt cracking on asphalt surfaced pavements) or for a range of values.

Prior to all interim and final submittals to the client project manager, DTS will perform a quality assurance review. A DTS representative will review all quality control memoranda and checks with the project manager and project quality control officer. This step assures the City of Cheyenne that DTS has followed all stated quality control procedures outlined in this response and the DTS Quality Assurance/Quality Control manual. DTS will update City Staff on any sections of roadway that could not be collected and the reasoning such as it is a private gated community, dirt roads, etc.

Data Reduction and Processing Checks

Pavement condition data collection crews typically perform a series of data checks in the field; once submitted, the pavement condition data is reviewed in the office for accuracy and completeness. In the office, the pavement condition data and images are processed according to standard procedures and analyzed for quality issues. After data collection and determination of condition ratings are complete, the final database is compiled and segmented according to the agency's location referencing system (LRS). At this time, segment lengths are checked against the master routing file to look for any missing segments. Final checks of the database typically include verifying proper format, checking for missing data and screening the entire database for errors.

The DTS team also adheres to the AASHTO adopted standards of practice for collecting network-level IRI, rut depth, cracking in asphalt pavements and faulting in jointed concrete pavements. The purpose of these standards was to help produce consistent results for use in network-level pavement management. Moreover, DTS equipment operates Texas A&M Transportation Institute certified profilers.

Before delivery, a series of quality control checks are conducted on the compiled database. Final checks of the database typically include verifying proper format, checking for missing data and screening the entire database for errors. This may involve logic checks and identifying data patterns such as a consecutive series of zeroes, null, repeating or error values or random values that are much larger or smaller than expected (i.e., out of range).

EXHIBIT B

Attached to Professional Services Agreement between the City of Cheyenne, Wyoming and Data Transfer Solutions, LLC for the Pavement Condition Survey Project

EXHIBIT B

City of Cheyenne, Wyoming Pavement Condition Survey Project Data Transfer Solutions, LLC				
Task	Description	Units	Unit Cost	Fee
1	Centerline Review, Route Planning and Mobilization	1	\$ 7,200.00	\$ 7,200.00
2a	Mobile Image Data Collection of Pavement and ROW Imaging (Units = Centerline Miles)	337	\$ 120.00	\$ 40,440.00
2b	Pavement Condition Index per the ASTM D6433 including Width (Units = Centerline Miles)	337	\$ 140.00	\$ 47,180.00
3	Import to MicroPAVER and Training	1	\$17,500.00	\$ 17,500.00
4	Pavement Condition Final Report	1	\$12,500.00	\$ 12,500.00
5	Quality Assurance/Quality Control (No Costing)			\$ -
				\$ -
				\$ -
Based on 337 Centerline Miles				
Fee Total				\$ 124,820.00

*DTS will invoice monthly based on percentage of work completed on each Task listed above.

EXHIBIT C

Attached to Professional Services Agreement between the City of Cheyenne, Wyoming and Data Transfer Solutions, LLC for the Pavement Condition Survey Project

EXHIBIT C

Proposed Schedule (City of Cheyenne) – complete in 120 days from NTP on July 2, 2018	
Task	Timeline
Centerline identification and mobile asset collection pre-planning	July 3, 2018
Project kickoff meeting	July 10, 2018
Mobile data collection (Pavement and ROW imagery)	7/10-8/31
Pavement condition rating per the ASTM D6433 (concurrent with collection)	7/17-9/30
Import to MicroPAVER and Training	October 30, 2018
Pavement Final Report	October 30, 2018